

Alternating currents

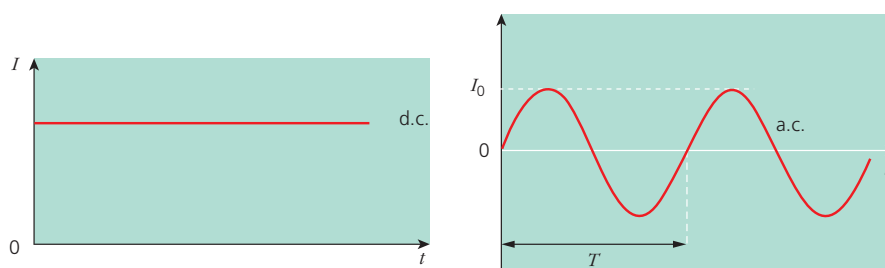
A Level Physics

Alternating current basics

An **alternating current** 交流电 (a.c.) keeps reversing direction. Mains supply is sinusoidal a.c.: I or V follows a sine wave in time:

$$I = I_0 \sin(\omega t), \quad V = V_0 \sin(\omega t).$$

(For a purely resistive load the **voltage** 电压 and **current** 电流 are in phase, which is the case in this syllabus.)



A steady direct current compared with a sinusoidal alternating current of peak I and period T

Key terms

- **period** 周期 T —the time for one full cycle. Unit: s.
- **frequency** 频率 f —cycles per second; $f = 1/T$. Mains is often 50 Hz or 60 Hz.
- **angular frequency** 角频率 $\omega = 2\pi f = 2\pi/T$.
- **peak value** 峰值 I_0 or V_0 —the largest value in a cycle (also called the amplitude).
- **peak-to-peak value** 峰峰值 $2I_0$ —from $+I_0$ to $-I_0$. Useful when reading an oscilloscope.

Reading a CRO trace

Same as for any wave (Topic 7), using a **cathode-ray oscilloscope** 示波器:

- horizontal divisions $\times$ **time-base** 时基 $\rightarrow$ period T , so $f = 1/T$.
- vertical divisions $\times$ y -gain $\rightarrow$ peak voltage V_0 (measure centre to peak, or peak-to-peak then halve).

Power delivered to a resistor

For a resistive load R , the instant **power** 功率 is $P(t) = I(t)^2 R$. With $I = I_0 \sin(\omega t)$:

$$P(t) = I_0^2 R \sin^2(\omega t).$$

This is always positive, with peak $I_0^2 R$ and minimum zero, oscillating at twice the frequency of I . The mean of $\sin^2(\omega t)$ over a cycle is $\frac{1}{2}$, so the **average power** is

$$\langle P \rangle = \frac{1}{2} I_0^2 R = \frac{1}{2} P_{\text{peak}}.$$

Average a.c. power in a resistor is **half the peak power**.

Root-mean-square (r.m.s.) values

The r.m.s. current $I_{\text{r.m.s.}}$ is the **steady direct current** that would give the same average power in the same **resistance** 电阻 R . From $\langle P \rangle = I_{\text{r.m.s.}}^2 R = \frac{1}{2} I_0^2 R$:

$$I_{\text{r.m.s.}} = \frac{I_0}{\sqrt{2}}, \quad V_{\text{r.m.s.}} = \frac{V_0}{\sqrt{2}}.$$

The $\sqrt{2}$ comes from the name **root-mean-square** 均方根: $I_{\text{r.m.s.}} = \sqrt{\langle I^2 \rangle}$ and $\langle \sin^2 \rangle = \frac{1}{2}$. (Only the sinusoidal case is needed.)

Why r.m.s. matters

Quoted a.c. values are r.m.s. values. "230 V mains" means $V_{\text{r.m.s.}} = 230$ V, with peak $V_0 = 230\sqrt{2} \approx 325$ V. Components must be rated for the peak, not the r.m.s. Average power then takes the d.c. form:

$$\langle P \rangle = I_{\text{r.m.s.}}^2 R = V_{\text{r.m.s.}}^2 / R = V_{\text{r.m.s.}} I_{\text{r.m.s.}}$$

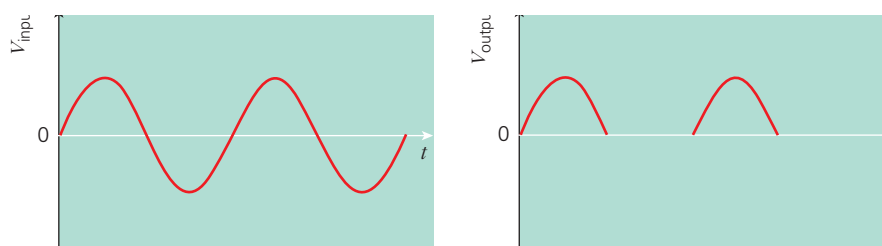
Rectification

Rectification 整流 turns an alternating voltage into a one-direction (d.c.-like) voltage, using **diodes** 二极管 (which conduct in only one direction).

Half-wave rectification

A **single diode** in series with the load passes only the positive half of each cycle; in the negative half the diode is **reverse-biased** 反向偏置 and no current flows. This is **half-wave rectification** 半波整流.

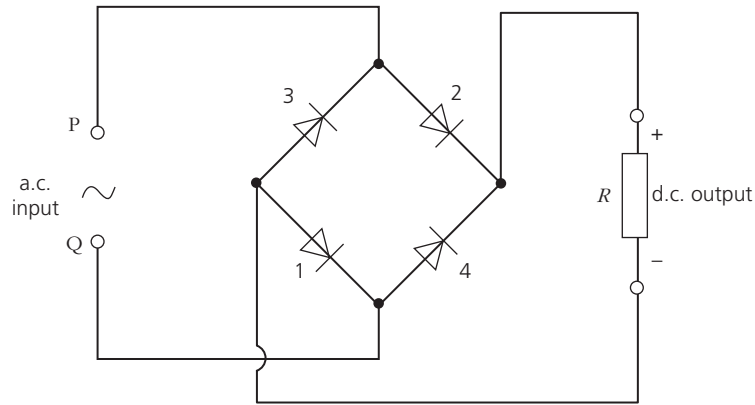
Output: positive half-waves with flat zero gaps. The mean output is $V_0/\pi \approx 0.32V_0$. Drawback: half the input is wasted and the output is very uneven.



In half-wave rectification a single diode passes only the positive half-cycles

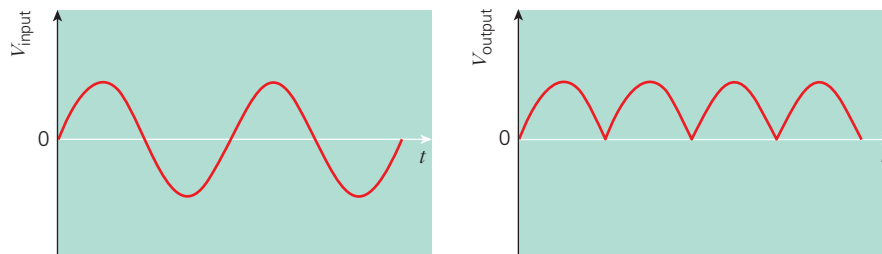
Full-wave rectification (bridge rectifier)

A **bridge rectifier** 桥式整流器 uses **four diodes** arranged so the current through the load always flows the same way, whichever a.c. terminal is positive — **full-wave rectification** 全波整流. On each half-cycle a different pair of diodes conducts, but the load always sees the same direction.



A four-diode bridge sends the load current the same way whichever a.c. terminal is positive

Output: a continuous run of positive half-waves (no gaps), at **twice** the input frequency. The mean output is $2V_0/\pi \approx 0.64V_0$ —double the half-wave value. It uses all the input and is smoother and easier to filter.



In full-wave rectification every half-cycle is used, giving a continuous run of positive humps

Drawing the diagrams

- half-wave: a.c. source —single diode —load R , in series.
- full-wave bridge: four diodes as the arms of a "diamond"; the a.c. input goes to one pair of opposite corners, the load R across the other pair. The diode directions make the load terminals keep the same polarity for either input polarity.

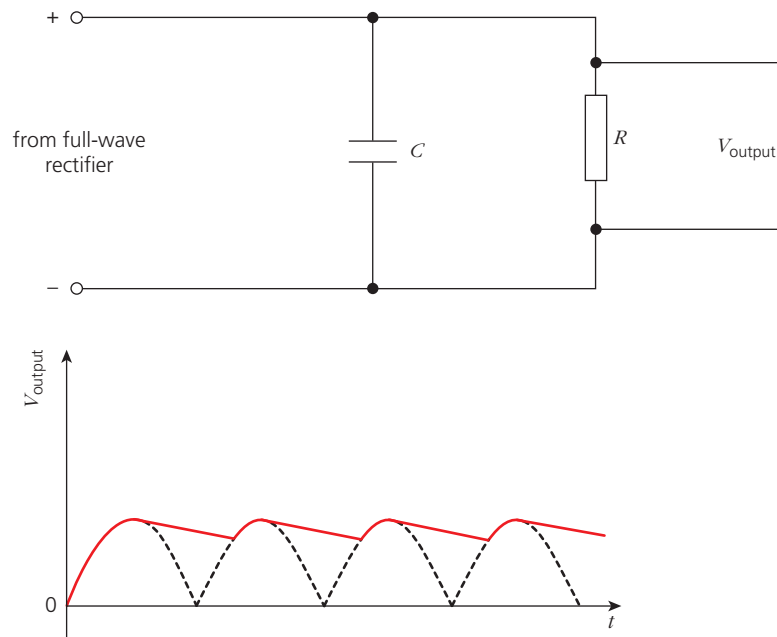
Smoothing with a capacitor

A rectifier's output is still bumpy. To smooth it, put a **capacitor** 电容器 C **in parallel with the load** R .

How it works

- on the **rising part** of each pulse, the capacitor charges up to near the peak.
- on the **falling part** (and any gap), the diodes are reverse-biased, so the capacitor **discharges through the load**, keeping current flowing. The voltage falls with **time constant** 时间常数 RC (Topic 19).
- at the next peak, the capacitor charges again, and the cycle repeats.

The output now sits near the peak with small dips. The size of the dips is the **ripple** 纹波 (this whole step is called **smoothing** 平滑).



A capacitor across the load smooths the rectified output, leaving only a small ripple

What reduces the ripple

- **larger C** $\rightarrow$ more stored charge $\rightarrow$ smaller dip between peaks $\rightarrow$ smaller ripple.
- **larger R** $\rightarrow$ smaller load current $\rightarrow$ slower discharge $\rightarrow$ smaller ripple.
- **higher rectified frequency** (full-wave is twice the input) $\rightarrow$ less time to discharge between peaks $\rightarrow$ smaller ripple.

In short, a large RC compared with the time between peaks gives a smoother output.

Purpose in summary

The smoothing capacitor reduces the ripple, giving a steadier d.c. voltage suitable for sensitive electronics.